IT Career Roadmap — One-List Master Plan (Certificate-First Edition)

Legend — Green = Free cert | Yellow = Paid, Necessary cert | Orange = Paid, Optional cert | Purple = Mixed/paired with a cert | Blue = Tool/account (not a course; no ce

Every course-type item below leads to SOME certificate/badge (free preferred). Pure tutorials with no credential path were replaced. Tools/accounts (GitHub, labs, VMs) are kept a

#	When	Program / Course to Complete (exact name)	Cost / Status	Role It Builds Toward
MONTH 0 — SETUP				
1	M0	Create a free GitHub account + a portfolio repository (tool/account — not a course)	Free	All Roles
2	M0	Install VirtualBox or VMware Workstation Player (home lab tool)	Free	All Roles
3	M0	Draft a baseline resume	Free	All Roles
MONTHS 1–3 — CORE FOUNDATIONS				
4	M1-3	Professor Messer — "CompTIA A+ Core 1 (220-1201)" free video course (free study — paired with the paid A+ exam below for the actual credential)	Free study	IT Support
5	M1-3	Professor Messer — "CompTIA A+ Core 2 (220-1202)" free video course; official A+ exam	Free study; exam ~$253–274/exam (~$506–548 total) — Paid, Necessary for the credential	IT Support
6	M1-3	Google IT Support Professional Certificate (Coursera) — issues a real Professional Certificate	$49/mo by default — apply for Coursera Financial Aid for $0	IT Support
7	M1-3	Cisco NetAcad — "Networking Basics" (22 hours, free self-paced; free NetAcad badge/certificate — Credly sync is not guaranteed)	Free (NetAcad badge/certificate)	Network Admin
8	M1-3	Cisco NetAcad — "Getting Started with Cisco Packet Tracer" (2 hours) — ✔️ COMPLETED Oct 4, 2026; certificate issued and downloadable from NetAcad	Free certificate (done)	Network Admin
9	M1-3	Cisco NetAcad — "Networking Devices and Initial Configuration" (22 hours; free, self-paced; continues Networking Basics)	Free (verify badge on course page)	Network Admin
10	M1-3	Cisco NetAcad — "Network Addressing and Basic Troubleshooting" (14 hours; free, self-paced)	Free (verify badge on course page)	Network Admin
11	M1-3	Cisco NetAcad — "Introduction to Cybersecurity" (6 hours; free, self-paced) — replaces a prior Juniper slot that had no standalone free certificate	Free (NetAcad badge/certificate)	Security / IT Support

#	When	Program / Course to Complete (exact name)	Cost / Status	Role It Builds Toward
12	M1-3	Cisco NetAcad — "Operating Systems Basics" (12 hours; free, self-paced; computer and mobile operating systems)	Free (verify badge on course page)	IT Support
13	M1-3	Cisco NetAcad — "Operating Systems Support" (10 hours; free, self-paced; troubleshooting Windows, macOS and mobile)	Free (verify badge on course page)	IT Support
14	M1-3	Microsoft Learn — "MD-100: Install the Windows client" (free training; feeds into the Applied Skills credential below — the module alone issues no certificate)	Free study (paired with credential below)	ADDS/Workstation
15	M1-3	Microsoft Learn — "MD-100: Support the Windows client environment" (free training; the actual current certification exam is MD-102, see Months 10-12)	Free study; MD-102 exam (if pursued) Paid, Optional — $165 confirmed	ADDS/Workstation
16	M1-3	Microsoft Learn — "Secure access to resources by using Microsoft Entra" (free training; feeds into the Applied Skills credential below)	Free study (paired with credential below)	ADDS/Workstation
17	M1-3	Set up a free Azure account (azure.microsoft.com/free) — auto-creates an Entra ID tenant (tool/account, required to complete the credential below; not a course itself)	Free (ID verification only, not charged)	ADDS/Workstation
18	M1-3	Microsoft Applied Skills — "Get started with identities and access using Microsoft Entra" — THE credential that rows 9, 11 and 12 build toward	Free credential	ADDS/Workstation
19	M1-3	Oracle MyLearn — "Become an OCI Foundations Associate 2026" (exam 1Z0-1085-26)	Free — genuinely free $0 exam, confirmed	DB Admin / Cloud
20	M1-3	freeCodeCamp — "Relational Database Certification" (genuine, verified, 100% free certification; hands-on SQL/PostgreSQL in a Linux terminal) — REPLACES PlanetScale's course, which teaches similar material but issues no certificate	Free certificate	DB Admin
21	M1-3	GitHub Skills — "Introduction to GitHub" (hands-on practice tool — profile achievement badges only, not a formal certificate; optional paid GitHub Foundations certification exists at ~$99 if you want a formal credential)	Free (no formal cert) / $99 optional cert	All Roles
✔	***M3***	***CHECKPOINT: Publish Portfolio Project #1 — a home-network diagram or a documented troubleshooting log — to your GitHub repo***	***Free***	***All Roles***
MONTHS 4–6 — SECURITY, IDENTITY & PRIMARY CLOUD				
22	M4-6	Fortinet Training Institute — "Fortinet Certified Fundamentals (FCF) in Cybersecurity"	Free badges	Security

#	When	Program / Course to Complete (exact name)	Cost / Status	Role It Builds Toward
23	M4-6	Palo Alto Networks Beacon — free "Cybersecurity Foundation" course (issues certificate of completion) — REPLACES Cyberpedia, which is reference content with no certificate	Free certificate	Security
24	M4-6	ISC2 — "Certified in Cybersecurity (CC)" self-paced course	Free study; exam now $199 + $50/yr (free 1MCC voucher closed to new applicants)	Security
25	M4-6	Microsoft Learn — "AZ-900: Microsoft Azure Fundamentals"	Free study; exam $99 — Paid, Optional	Cloud Engineer
26	M4-6	Microsoft Applied Skills — Azure task-based credentials	Free credential	Cloud Engineer
27	M4-6	AWS Skill Builder — "Cloud Practitioner Essentials" (issues a free digital course-completion certificate, separate from the paid AWS Certification exam)	Free completion certificate	Cloud Engineer
28	M4-6	Google Cloud Skills Boost — "Cloud Digital Leader" / "Cloud Infrastructure" intro (issues a free Skill Badge on completion)	Free skill badge	Cloud Engineer
29	M4-6	MongoDB University — "Introduction to MongoDB" (issues free completion certificate)	Free certificate	DB Admin
30	M4-6	Redis University — "Get Started with Redis" (issues free completion certificate)	Free certificate	DB Admin
✔	***M6***	***CHECKPOINT: Sit ONE certification — Oracle OCI Foundations ($0) OR Cisco CCST ($125) OR AZ-900 ($99)***	***Free–$125***	***All Roles***
MONTHS 7–9 — PICK ONE SPECIALIZATION TRACK				
Read all 5 tracks below (A–E), then complete ONLY the items under the ONE track that matches your target role. Skip the other four entirely.				
TRACK A — NETWORK / SYSADMIN (target: Network Admin, Systems Admin)				
31	M7-9	Cisco NetAcad — "Network Support and Security" (12 hours; free, self-paced; troubleshooting and user access control). CORRECTION: Cisco's CCNA courses (Introduction to Networks, Switching/Routing/Wireless, Enterprise Networking) are instructor-led only and need a class code from a school or academy — they are NOT free self-paced	Free (verify badge on course page)	Network Admin
32	M7-9	Juniper Open Learning — "JNCIA-Junos" (elective; training is free but issues no standalone certificate of its own — only the paid JNCIA exam does)	Free study only (no free cert)	Network Admin

#	When	Program / Course to Complete (exact name)	Cost / Status	Role It Builds Toward
33	M7-9	VMware Hands-on Labs — ESXi & vSphere Fundamentals (hands-on lab tool, kept for practical virtualization practice; no certificate offered)	Free (no cert)	SysAdmin
34	M7-9	Choose ONE certification exam: Cisco CCST Networking ($125) OR Cisco CCNA 200-301 ($300) OR CompTIA Network+ ($399) (CCNA prep needs outside study material, since the CCNA NetAcad courses are instructor-led)	Paid, Necessary — pick one, not all three	Network Admin
35	M7-9	CAPSTONE: Build a routed/switched network topology in Packet Tracer or GNS3; publish configs and troubleshooting notes to GitHub (your own deliverable, not a course)	Free	Network Admin
TRACK B — CLOUD / DEVOPS (target: Cloud Engineer, DevOps Engineer)				
36	M7-9	IBM SkillsBuild — "Containers & Kubernetes" fundamentals (free badge) — REPLACES Docker's own "Get Started" guide, which has no certificate mechanism	Free badge	DevOps/Cloud
37	M7-9	Linux Foundation — "Introduction to Kubernetes (LFS158)" via edX (free honor-code certificate available) — REPLACES kubernetes.io's own tutorials, which issue no certificate	Free certificate	DevOps/Cloud
38	M7-9	HashiCorp Developer — Terraform "Get Started" tutorials (free study; the real credential is HashiCorp Certified: Terraform Associate, a relatively low-cost paid exam)	Free study; Terraform Associate exam Paid, Optional (~$70.50)	DevOps/Cloud
39	M7-9	Linux Foundation — "Introduction to Linux (LFS101)" (free honor-code certificate)	Free certificate	DevOps/Cloud
40	M7-9	GitHub Skills — "GitHub Actions: Hello World" (hands-on practice tool, no formal certificate)	Free (no cert)	DevOps/Cloud
41	M7-9	Optional certification: Microsoft Applied Skills (Azure, free) OR AZ-104 ($165) OR AWS Cloud Practitioner ($100)	Free–$165 — Paid, Optional	DevOps/Cloud
42	M7-9	CAPSTONE: Containerize an app, deploy it through a CI/CD pipeline, provision infrastructure with Terraform, document rollback/monitoring (your own deliverable)	Free	DevOps/Cloud
TRACK C — DATABASE (target: DB Admin, Data Engineer)				
43	M7-9	freeCodeCamp — "Relational Database Certification" (deeper practice, reinforcing Phase 1's certificate)	Free certificate	DB Admin

#	When	Program / Course to Complete (exact name)	Cost / Status	Role It Builds Toward
44	M7-9	MongoDB University — "Aggregation Framework" course (free completion certificate)	Free certificate	DB Admin
45	M7-9	Couchbase Academy — free NoSQL courses (free completion certificate; Associate exam $50 if a target employer requires it)	Free certificate; $50 exam optional	DB Admin
46	M7-9	DataStax Academy — Cassandra Fundamentals (elective, employer-stack driven; likely issues a completion badge — exact current status not independently reverified this round)	Free (medium confidence on cert)	DB Admin
47	M7-9	CAPSTONE: Design a normalized schema, load sample data, write 10–15 real query/reporting scripts, publish to GitHub (your own deliverable)	Free	DB Admin
TRACK D — SECURITY & SOC (target: Security Analyst, SOC Analyst)				
48	M7-9	Fortinet Certified Fundamentals (FCF) — reinforce from Phase 2	Free badges	Security
49	M7-9	Cisco NetAcad — "Network Defense" (27 hours; free, self-paced; monitor and protect a network, evaluate security alerts)	Free (verify badge on course page)	Security/SOC
50	M7-9	Cisco NetAcad — "Cyber Threat Management" (16 hours; free, self-paced; cybersecurity governance and threat management)	Free (verify badge on course page)	Security/SOC
51	M7-9	Cisco NetAcad — "Data and Tools for Defense Analysts" (6 hours; free, self-paced; tools and data sources SOC analysts use)	Free (verify badge on course page)	Security/SOC
52	M7-9	Cisco NetAcad — "Introduction to Splunk" (1 hour; free, self-paced) — replaces the earlier vague Splunk Education entry	Free (verify badge on course page)	Security/SOC
53	M7-9	Cisco NetAcad — "SOC Essentials: Investigating with Splunk" (3 hours; free, self-paced)	Free (verify badge on course page)	Security/SOC
54	M7-9	Cisco NetAcad — "Ethical Hacker" (70 hours; free, self-paced; OPTIONAL — the longest free option, only if you want offensive-security depth)	Free (verify badge on course page)	Security/SOC
55	M7-9	Wazuh or Security Onion — install in a home VM for your SOC lab (hands-on lab tools, not courses; your home-lab evidence pairs with the certification below)	Free, self-hosted	Security/SOC

#	When	Program / Course to Complete (exact name)	Cost / Status	Role It Builds Toward
56	M7-9	Choose ONE certification: ISC2 Certified in Cybersecurity ($199+ $50/yr) OR CompTIA Security+ ($439)	Paid, Necessary — pick one, not both	Security/SOC
57	M7-9	CAPSTONE: Build a home SOC lab (Wazuh/Security Onion in a VM); simulate and document an incident response (your own deliverable)	Free	Security/SOC
TRACK E — AI / ML (target: AI Specialist, Prompt Engineer)				
58	M7-9	Google Cloud Skills Boost — "Introduction to Generative AI" / "Generative AI Fundamentals" skill badge	Free skill badge	AI Engineer
59	M7-9	IBM SkillsBuild — "AI Fundamentals: Foundations for Understanding AI" (9 hours; free, self-paced; machine learning, neural networks, deep learning, generative AI)	Free (verify badge on course page)	AI Engineer
60	M7-9	IBM SkillsBuild — "AI Fundamentals: Language and Vision in AI" (10 hours; free, self-paced)	Free (verify badge on course page)	AI Engineer
61	M7-9	Cisco NetAcad — "Core Generative AI Skills" (7 hours; free, self-paced; first course of Cisco's AI Business Practitioner collection)	Free (verify badge on course page)	AI Engineer
62	M7-9	Cisco NetAcad — "Prompt Like an Engineer" (9 hours; free, self-paced; design, iterate and verify prompts, local and cloud LLMs)	Free (verify badge on course page)	AI Engineer
63	M7-9	Hugging Face — "Agents Course" at huggingface.co/agents-course (free certificate)	Free certificate	AI/ML Developer
64	M7-9	NVIDIA Deep Learning Institute — free self-paced courses (several issue a certificate of competency on passing the in-course assessment — confirm per course)	Free (verify per-course cert status)	AI Engineer
65	M7-9	AWS AI Practitioner learning path (issues a free digital course-completion certificate, separate from the paid AWS exam)	Free completion certificate	AI Engineer
66	M7-9	Optional certification: Microsoft AI-901 ($99) OR AWS Certified AI Practitioner exam ($100)	Paid, Optional	AI Engineer
67	M7-9	CAPSTONE: Build and deploy a small AI-powered app (e.g. support-ticket summarizer) using a hosted LLM API; document architecture and safety notes (your own deliverable)	Free	AI Engineer
✔	***M9***	***CHECKPOINT: Confirm you completed only ONE track above, and that your capstone project is published to GitHub***	***Free***	***Varies by track***

#	When	Program / Course to Complete (exact name)	Cost / Status	Role It Builds Toward
MONTHS 10–12 — ENTERPRISE TOOLS & JOB SEARCH				
68	M10-12	Microsoft Learn — "Prepare infrastructure for devices using Microsoft Intune and Microsoft Entra ID" (free training; feeds toward Intune-related Applied Skills credentials and/or MD-102)	Free study (paired with credentials above)	IT Specialist
ELECTIVE — ENDPOINT & MOBILE DEVICE MANAGEMENT (for IT Support/Helpdesk roles requiring M365, Intune, Defender, Apple & Android MDM)				
69	M10-12	Microsoft Applied Skills — "Defend against cyberthreats with Microsoft Defender XDR" (hands-on lab assessment covering endpoint detection/response)	Free credential	IT Support / Security
70	M10-12	Microsoft Learn — "App management using Microsoft Endpoint Manager" (free training covering Intune app management, MAM, BYOD, and app protection policies)	Free study	IT Support
71	M10-12	Jamf 100 Course (jamf.com/training/100) — free training covering macOS/iOS fundamentals and Apple device management (MDM) via Jamf Pro	Free study; optional "Jamf Certified Associate" exam Paid, Optional ($100)	IT Support
72	M10-12	Android Enterprise Academy (androidenterprise.training) — free courses covering fleet-level Android device management	Free — training AND the "Android Enterprise Associate" certification are both genuinely free	IT Support
73	M10-12	Cisco NetAcad — "Endpoint Security" (27 hours; free, self-paced; securing the network "all the way to the edge")	Free (verify badge on course page)	IT Support / Security
ELECTIVE — VIRTUALIZATION, STORAGE & BACKUP (for Technical Support Engineer roles: Red Hat/OpenShift, VMware/Hyper-V, SAN storage, enterprise backup)				
74	M10-12	NDG "Linux Unhatched" (8 hours; free, self-paced, hosted on Cisco NetAcad); go on to NDG "Linux Essentials" (70 hours, also free) for depth — it prepares you for the LPI Linux Essentials exam, which is a separate paid credential	Free (verify badge on course page); LPI exam paid	Virtualization / Support Engineer
75	M10-12	Red Hat — "Red Hat OpenShift Virtualization Technical Overview (DO016)" (free on-demand video course, no prerequisites; covers creating/managing VMs, networking, storage, HA and VM migration on OpenShift)	Free study (certificate status not confirmed); formal EX316 credential is Paid, Optional — exam price not confirmed	Virtualization / Support Engineer
76	M10-12	Red Hat — "Containers, Kubernetes and Red Hat OpenShift Technical Overview (DO080)" (free course; recommended before OpenShift Virtualization)	Free study (certificate status not confirmed)	Virtualization / Support Engineer
77	M10-12	Veeam University Free — free on-demand training on Veeam Backup & Replication / Veeam Data Platform (issues a FREE certificate on course completion; the formal VMCE/VMCA exams require a paid instructor-led class)	Free certificate	Backup / Support Engineer

#	When	Program / Course to Complete (exact name)	Cost / Status	Role It Builds Toward
78	M10-12	Veeam Community Edition (free license, up to 10 workloads) — install in a home-lab VM so you can practice backup jobs and restores (lab tool, not a course)	Free (no cert)	Backup / Support Engineer
79	M10-12	Dell Technologies — "Information Storage and Management Foundations" (SAN/storage credential). NOTE: no free SAN/storage certificate was found — this is the credible paid option; exam fee not independently confirmed this round	Paid, Optional — price not confirmed; no free equivalent found	Storage / Support Engineer
80	M10-12	Simplilearn SkillUp — "Introduction to ITIL Foundation 4" (free course + free completion certificate; official ITIL exam separate and paid)	Free certificate	IT Specialist
81	M10-12	Atlassian University — Jira Service Management fundamentals (several free Atlassian University courses issue a completion certificate — confirm the specific course you take)	Free (verify per-course cert status)	IT Support
82	M10-12	Sit the 1–2 paid exams that matter most for your chosen track (see Track section above); if targeting broader identity/cloud admin work, consider SC-300: Identity and Access Administrator ($165)	Paid — role-dependent	All Roles
83	M10-12	Finalize GitHub portfolio (2+ projects) + tailor resume per target role	Free	Career Expansion
84	M10-12	Free mock-interview practice; apply to 5–10 targeted roles/week	Free	Career Expansion
MONTHS 13–15 — BUFFER (OPTIONAL)				
85	M13-15	Retake any failed certification exam, or start a second specialization track	Varies	All Roles